Second Partial

Fco. Vazquez

Tec de Monterrey

Rule of derivatives

Power rule

$$\frac{\mathrm{d}}{\mathrm{d}x}x^n = nx^{n-1}$$

Chain rule

$$\frac{\mathrm{d}}{\mathrm{d}x}u(v(x)) = u'(v(x)) \cdot v'(x)$$

Quotient rule

$$\frac{\mathrm{d}}{\mathrm{d}x}\frac{u(x)}{v(x)} = \frac{u'v - uv'}{v^2}$$

Product rule

$$\frac{\mathrm{d}}{\mathrm{d}x}u(x)v(x) = u'v + uv'$$

Example: $f(x) = \frac{x}{\left(x^2+1\right)^{3/2}}$

Quotient rule

The function $f(x)$ is a quotient of functions, where $u = x$ and $v = \left(x^2+1\right)^{3/2}$. By applying the quotiente rule, $\frac{\mathrm{d}}{\mathrm{d}x} u/v = \frac{1}{v^2}(u'v - uv')$, we get,

$$
\begin{aligned}
f'(x) &= \frac{1}{\left[\left(x^2+1\right)^{3/2}\right]^2} \left[\left(\frac{\mathrm{d}}{\mathrm{d}x} x\right)\left(x^2+1\right)^{3/2} - x\left(\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2+1\right)^{3/2}\right)\right] \\
&= \frac{1}{\left(x^2+1\right)^3} \left[\left(\frac{\mathrm{d}}{\mathrm{d}x} x\right)\left(x^2+1\right)^{3/2} - x\left(\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2+1\right)^{3/2}\right)\right]
\end{aligned}
$$

Example: $f(x) = \frac{x}{\left(x^2+1\right)^{3/2}}$

$$f'(x) = \frac{1}{\left(x^2+1\right)^3}\left[\left(\frac{\mathrm{d}}{\mathrm{d}x}x\right)\left(x^2+1\right)^{3/2} - x\left(\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2+1\right)^{3/2}\right)\right]$$

Power rule: $\frac{\mathrm{d}}{\mathrm{d}x}x$

$$\frac{\mathrm{d}}{\mathrm{d}x}x = 1x^{1-1} = 1x^0 = 1 \cdot 1 = 1$$

Example: $f(x) = \frac{x}{(x^2+1)^{3/2}}$

Replace

$$
\begin{aligned}
f'(x) &= \frac{1}{(x^2+1)^3}\left[\cancelto{1}{\left(\frac{\mathrm{d}}{\mathrm{d}x}x\right)}\left(x^2+1\right)^{3/2} - x\left(\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2+1\right)^{3/2}\right)\right] \\
&= \frac{1}{(x^2+1)^3}\left[1\cdot\left(x^2+1\right)^{3/2} - x\left(\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2+1\right)^{3/2}\right)\right]
\end{aligned}
$$

Example: $f(x) = \frac{x}{\left(x^2+1\right)^{3/2}}$

Chain rule: $\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2+1\right)^{3/2}$

Here, $u = x^{3/2}$ and $v = x^2+1$, so that $(u \circ v)(x) = \left(x^2+1\right)^{3/2}$. With this, we can apply the chain rule, $\frac{\mathrm{d}}{\mathrm{d}x} u(v(x)) = u'(v(x)) \cdot v'(x)$,

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2+1\right)^{3/2} &= \left.\frac{\mathrm{d}}{\mathrm{d}x} x^{3/2}\right|_{x=x^2+1} \cdot \left[\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2+1\right)\right] \\
&= \left.\frac{3}{2} x^{1/2}\right|_{x=x^2+1} \cdot [2x+0]
\end{aligned}$$

Example: $f(x) = \frac{x}{\left(x^2+1\right)^{3/2}}$

Chain rule: $\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2+1\right)^{3/2}$

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2+1\right)^{3/2} &= \frac{3}{2}x^{1/2}\bigg|_{x=x^2+1} \cdot [2x+0] \\ &= \frac{3}{2}\sqrt{x^2+1} \cdot 2x \\ &= 3x\sqrt{x^2+1}\end{aligned}$$

$$\textit{Example: } f(x) = \frac{x}{\left(x^2+1\right)^{3/2}}$$

Replace

$$f'(x) = \frac{1}{\left(x^2+1\right)^3}\left[1\cdot\left(x^2+1\right)^{3/2} - x\cancelto{3x\sqrt{x^2+1}}{\left(\frac{\mathrm{d}}{\mathrm{d}x}\left(x^2+1\right)^{3/2}\right)}\right]$$

$$= \frac{1}{\left(x^2+1\right)^3}\left[1\cdot\left(x^2+1\right)^{3/2} - x\left(3x\sqrt{x^2+1}\right)\right]$$

Example: $f(x) = \frac{x}{\left(x^2+1\right)^{3/2}}$

Simplify

$$f'(x) = \frac{1}{\left(x^2+1\right)^3}\left[1\cdot\left(x^2+1\right)^{3/2} - x\left(3x\sqrt{x^2+1}\right)\right]$$

$$= \frac{1}{\left(x^2+1\right)^3}\left[\left(x^2+1\right)^{3/2} - 3x^2\sqrt{x^2+1}\right]$$

$$\boxed{f'(x) = \frac{1-2x^2}{\sqrt{\left(x^2+1\right)^5}}}$$

Class Activity

Compute the derivative of the following functions:

$$f(x) = \frac{2}{1+e^{-2(x-4)}} \qquad g(x) = \frac{2x}{\left(x^2+4\right)^{3/2}}$$

$$f'(x) = \frac{4e^{-2(x-4)}}{\left[1+e^{-2(x-4)}\right]^2} \qquad g'(x) = \frac{8-4x^2}{\sqrt{\left(x^2+4\right)^5}}$$

Power rule: $\frac{\mathrm{d}}{\mathrm{d}x}x^n = nx^{n-1}$ Chain rule: $\frac{\mathrm{d}}{\mathrm{d}x}u(v(x)) = u'(v(x)) \cdot v'(x)$

Quotient rule: $\frac{\mathrm{d}}{\mathrm{d}x}\frac{u(x)}{v(x)} = \frac{u'v - uv'}{v^2}$ Product rule: $\frac{\mathrm{d}}{\mathrm{d}x}u(x)v(x) = u'v + uv'$